18 A metal wire has length 5.2 m and diameter 1.0 mm. The metal has Young modulus 360 GPa.

The wire is fixed at one end and a force is applied to the other end. The force extends the wire by 7.2 mm. The wire obeys Hooke's law.

What is the force applied to the wire?

A 1.2×10^2 N

B 3.9×10^2 N

C 5.0×10^2 N

D 1.6×10^3 N